

Information Technology Institute (ITI)
Analog IC Design

Lab 07

Five-Transistor OTA Design

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1 Part 1: OTA Design (Hand Analysis)

Specs recap

Parameter	Value
Supply VDD	1.8 V
Load C_L	5 pF
Open-loop DC gain	≥ 34 dB
CMRR @ DC	≥ 74 dB
Phase margin	$\geq 70^\circ$
Reference (seed) current	10 μ A
CMIR low	≤ 1.2 V
CMIR high	≥ 1.5 V
GBW	≥ 5 MHz
Current consumption / Area	Minimize

Q: NMOS or PMOS?

A: NMOS, since the CMIR is closer to VDD we will choose nmos.

Bias current – simple approach

The dominant pole is at the output node, so $GBW = g_{m1}/(2\pi C_L)$:

$$g_{m1} \geq 2\pi \cdot GBW \cdot C_L = 2\pi(5 \text{ MHz})(5 \text{ pF}) = 157.1 \mu\text{A/V}$$

Assume $g_m/I_D|_1 = 15$ as a first guess (moderate inversion):

$$I_{D1} = \frac{g_{m1}}{g_m/I_D} = \frac{157.1}{15} = 10.47 \mu\text{A}$$

so ID1 will be 10 μ A

$$g_m/I_D|_1 = \frac{157.1}{10} = 15.7 \text{ S/A}$$

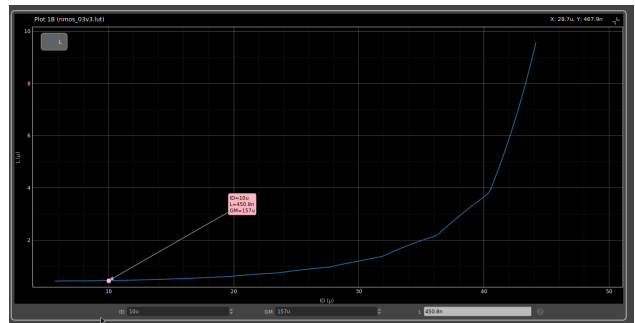
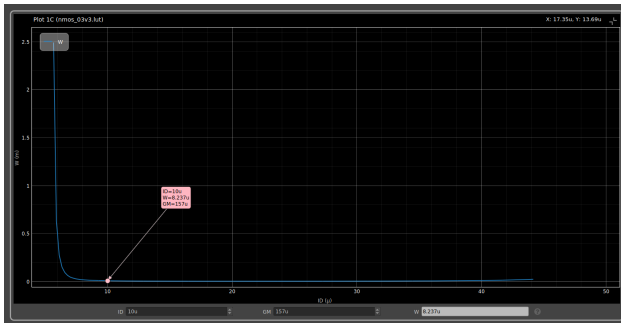
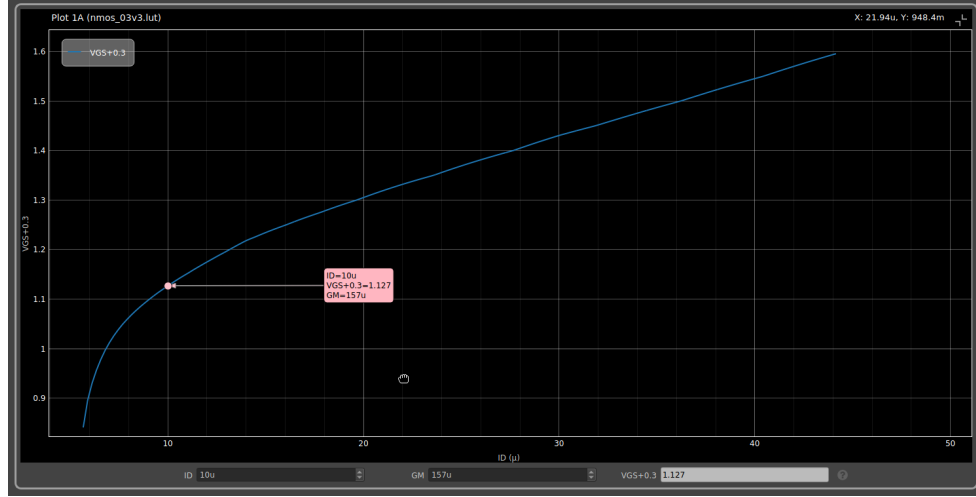


Figure 1: W vs I_D (left) and L vs I_D (right)

assume the tail has $V_{DS,tail} = 0.3$ V, and plot $V_{icm,min} = V_{GS1} + 0.3$ vs ID:

Figure 2: $V_{icm,min} = V_{GS1} + 0.3 \text{ V}$ vs I_D

At $I_D = 10 \mu\text{A}$: $V_{icm,min} = 1.127 \text{ V}$, which is below the 1.2 V .

Gain spec → sizing M3/M4 (load)

$$A_{v0} = g_{m1}(r_{o2} \parallel r_{o4}) = \frac{g_{m1}}{2\lambda_L I_D} \geq 50.1 \text{ V/V (34 dB)}$$

$$\lambda_L \leq \frac{g_{m1}}{2I_D \cdot 50.1} = \frac{157.1\mu}{2(10\mu)(50.1)} = 0.157 \text{ V}^{-1}, r_o = 634 \text{ kohm}$$

picking $g_m/I_D|_{3,4} = 10 \text{ S/A}$ for the load devices. Using CMIR-Max:

$$CMIR_{MAX} = 1.5 = V_{DD} + V_{gs1,2} - V_{dsat1,2} - V_{gs3,4} \text{ (Using Value of } V_{th2} \text{ from ADT)}$$

$$\rightarrow V_{gs3,4} = 1.023 \approx 1 \text{ V}$$

CMRR spec → sizing M5 (tail)

$$CMRR_{spec} - A_{v0,spec} = 74 - 34 = 40 \text{ dB}$$

$$CMRR = A_{v0} + 20 \log_{10}(2g_{m3}r_{o5}) \Rightarrow 2g_{m3}r_{o5} = 100 \Rightarrow g_{m3}r_{o5} = 50$$

With $g_m/I_D|_3 = 10 \Rightarrow g_{m3} = 100 \mu\text{A/V}$:

$$r_{o5} = \frac{50}{100 \mu\text{A/V}} = 500 \text{ k}\Omega \Rightarrow \lambda_5 = \frac{1}{r_{o5}I_5} = \frac{1}{(500\text{k})(20\mu)} = 0.1 \text{ V}^{-1}$$

assumed $V_{DS,tail} = 0.3 \text{ V}$, $g_m/I_D = 10$ for the tail.

Final sizing table

Table 1: Final transistor sizes (from ADT)

Device	W (μm)	L (nm)	I_D (μA)	g_m ($\mu\text{A/V}$)	g_m/I_D (S/A)	V^* (V)	V_{ov} (V)	V_{dsat} (V)
M1, M2	10	450	10	157	15.76	0.128	0.031	0.104
M3, M4	7.5	450	10	100.0	9.9	0.201	0.156	0.174
M5	2×4.2	900	19.70	199.0	10	0.202	0.147	0.173
M6	4.2	900	10.0	100.8	9.86	0.202	0.147	0.173

Parameter settings for a tail/reference device:

- ID: 10u
- gm/ID: 10
- L: 0.9u
- VDS: 0.3
- VSB: 0
- Stack: 1

Results: TT-27.0

	Name	Value
1	ID	10u
2	IG	N/A
3	L	900n
4	W	4.2u
5	VDB	300m
6	VGB	837.5m
7	VGS	837.5m
8	VDS	300m
9	VSB	0
10	VGB	837.5m
11	VDB	300m
12	gm/ID	9.861
13	Vstar	202.8m

Figure 3: Tail / reference device (M5, M6) sizing

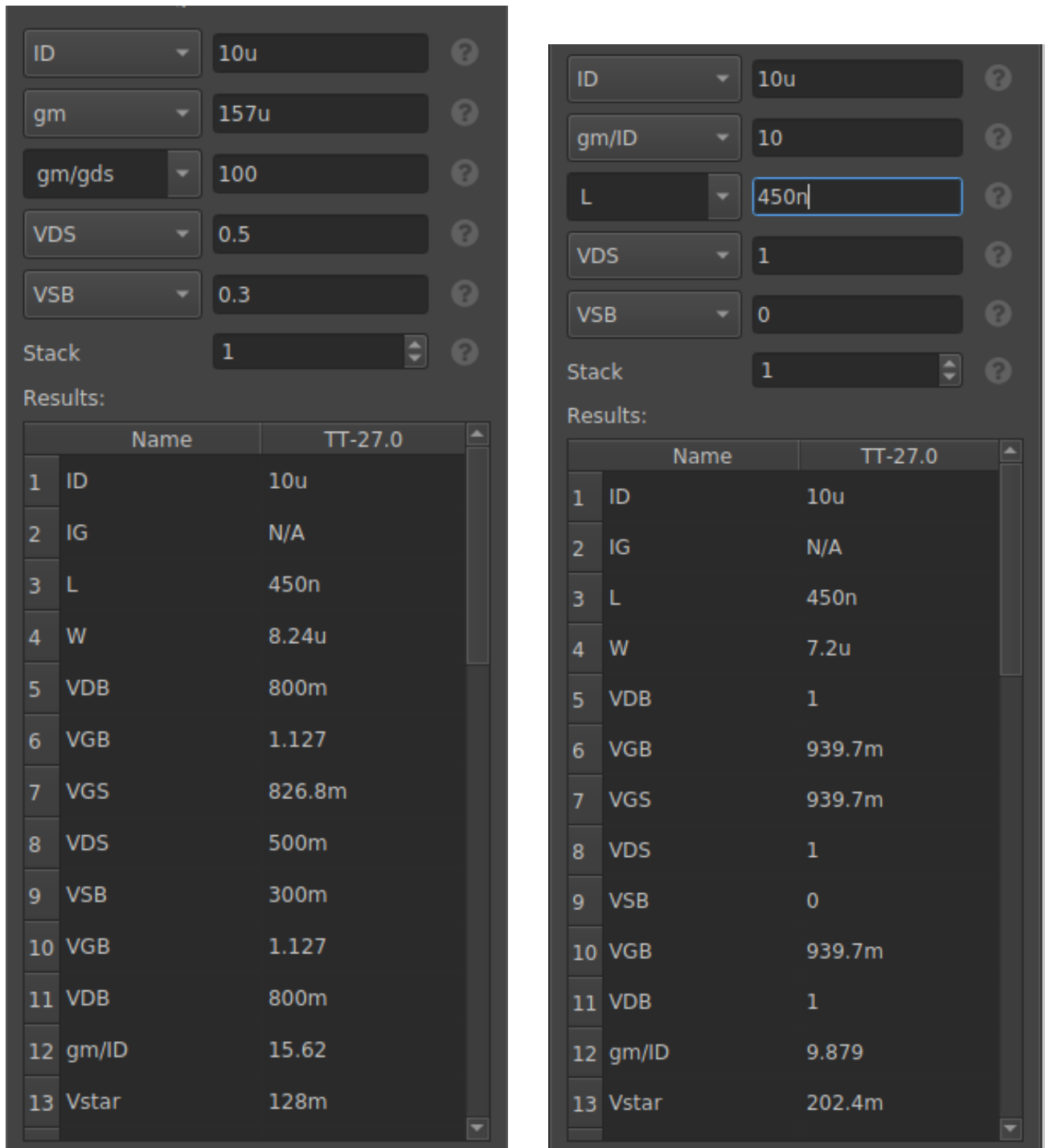


Figure 4: Input pair (M1/M2) sizing (left) and Load (M3/M4) sizing (right)

The width of M1,2 is 8.5u but can be increased to 9.5u to achieve GBW=5MHz.

2 Part 2: Open-Loop OTA Simulation

2.1 DC operating point

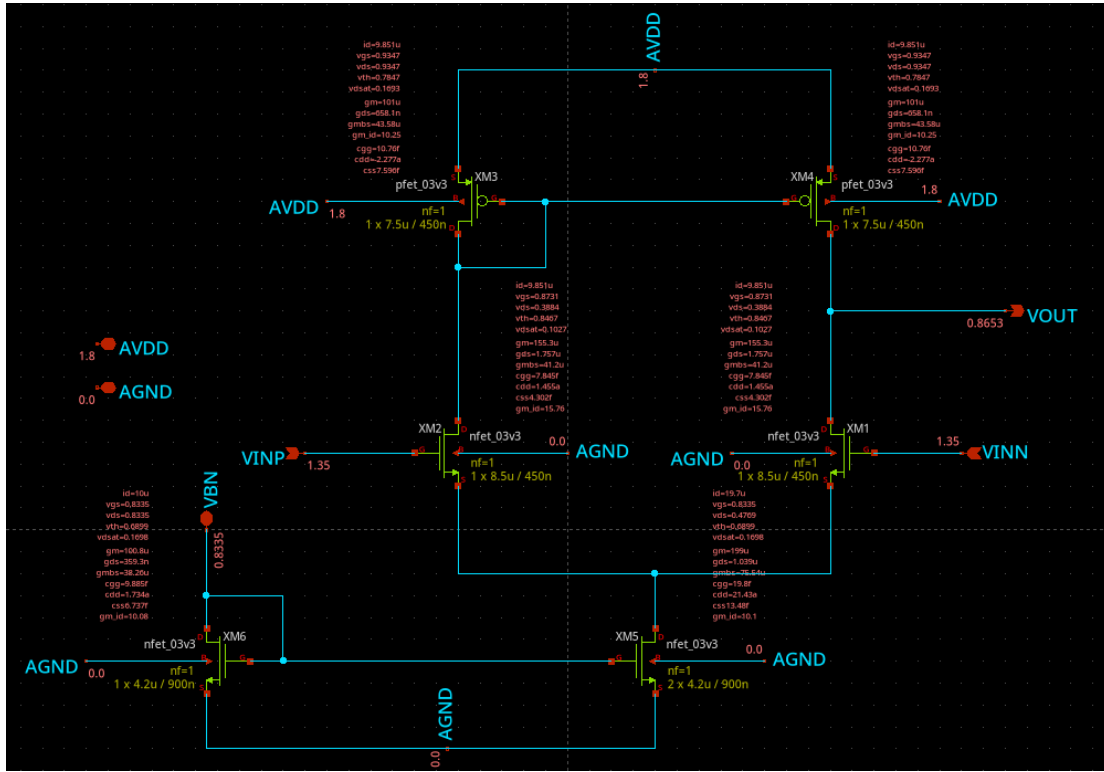


Figure 5: Open-loop schematic with DC node voltages annotated ($V_{ICM} = 1.35\text{ V}$)

Q: Is the current and gm in the input pair exactly equal?

A: Yes, because the circuit is perfectly symmetric: both inputs have the same common-mode signal there's no feedback to introduce any imbalance.

```

No. of Data Rows : 1
binary raw file "ota_testbench.raw"
=== MOSFET Operating Points ===
BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m,x1,xm6,m0      m,x1,xm5,m0      m,x1,xm4,m0
model      nfet_03v3,9      nfet_03v3,9      pfet_03v3,8
id          1e-05          1.97023e-05      9.85119e-06
vgs         0.833472      0.833472      0.934708
vds         0.833469      0.47688      0.934707
vth         0.689898      0.689901      0.784687
vdsat       0.16984      0.169838      0.169334
gm          0.000100779      0.00019904      0.000100962
gds         3.59306e-07      1.0389e-06      6.58111e-07
gmb         3.82588e-05      7.55337e-05      4.35795e-05

BSIM4v5: Berkeley Short Channel IGFET Model-4
device      m,x1,xm3,m0      m,x1,xm2,m0      m,x1,xm1,m0
model      pfet_03v3,8      nfet_03v3,8      nfet_03v3,8
id          9.85119e-06      9.85116e-06      9.85116e-06
vgs         0.934708      0.873113      0.873113
vds         0.934707      0.388401      0.388401
vth         0.784687      0.846704      0.846704
vdsat       0.169334      0.102713      0.102713
gm          0.000100962      0.000155262      0.000155262
gds         6.58111e-07      1.75723e-06      1.75723e-06
gmb         4.35795e-05      4.12003e-05      4.12003e-05

=== Bias Points ===
M1: gm/id = 15.7608 S/A, Vstar = 0.126897 V
M2: gm/id = 15.7608 S/A, Vstar = 0.126897 V
M3: gm/id = 10.2487 S/A, Vstar = 0.195146 V
M4: gm/id = 10.2487 S/A, Vstar = 0.195146 V
M5: gm/id = 10.1024 S/A, Vstar = 0.197974 V
M6: gm/id = 10.0778 S/A, Vstar = 0.198455 V
ngspice 1 ->

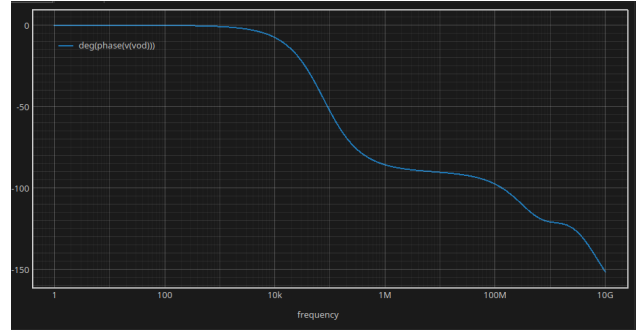
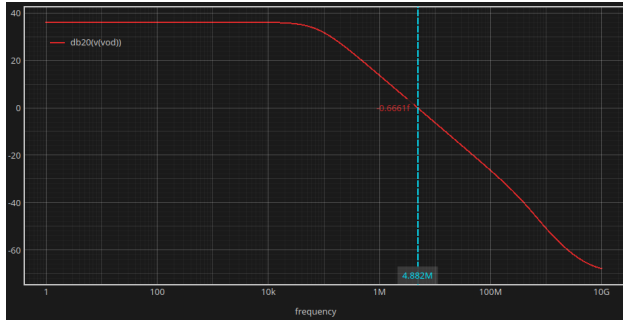
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Figure 6: Ngspice op-point

Q: What is the DC voltage at VOUT? Why?

A: $V_{OUT} = 0.8653$ V. it follows $V_F = V_{DD} - V_{SG3} = 1.8 - 0.9347 = 0.8653$ V, since the circuit is symmetric.

2.2 Differential small-signal AC

Figure 7: $|A_{v,diff}|$ vs frequency (dB) (left) and $\angle A_{v,diff}$ vs frequency (right)

($g_{m1} = 155.26 \mu A/V$, $g_{ds2} = 1.757 \mu A/V$, $g_{ds4} = 0.658 \mu A/V$):

$$A_{v,diff} = \frac{g_{m1}}{g_{ds2} + g_{ds4}} = \frac{155.26}{2.415} = 64.3 \text{ V/V} = 36.16 \text{ dB}$$

$$f_{p,out} = \frac{g_{ds2} + g_{ds4}}{2\pi C_L} = \frac{2.415 \mu A/V}{2\pi(5 \text{ pF})} \approx 76.9 \text{ kHz}$$

$$GBW = \frac{g_{m1}}{2\pi C_L} = \frac{155.26 \mu A/V}{2\pi(5 \text{ pF})} \approx 4.94 \text{ MHz}$$

Since it's a single dominant pole, PM should be close to 90° .

Table 2: Differential AC – hand calc vs sim

Quantity	Hand calc	Simulation
$A_{v,diff}$	36.16 dB	36.13 dB
BW_{diff}	76.9 kHz	76.1 kHz
GBW	4.94 MHz	4.877 MHz
PM	$\approx 90^\circ$	90.51°

```

Using SPARSE 1.3 as Direct Linear Solver
No. of Data Rows : 101
av_diff_db      = 3.613337e+01
av_diff_mag     = 6.407206e+01
bw_diff         = 7.612231e+04
ugf             = 4.894338e+06
p_ugf           = -8.948741e+01
av_cm_db        = -4.797362e+01
av_cm_mag       = 3.993180e-03
bw_cm           = 7.613349e+04
=== AC Results ===
Av Diff: 36.1334 dB, BW: 76122.3 Hz
UGF: 4.89434E+06 Hz
GBW: 4.87731E+06 Hz
PM: 90.5126 deg
Av CM: -47.9736 dB, BW: 76133.5 Hz
CMRR: 84.107 dB
binary raw file "ota_ol.raw"
ngspice 1 ->

```

Figure 8: Ngspice AC

To obtain GBW larger we can increase the width of M1,2 to be 9.5u instead of 8.5u

```

No. of Data Rows : 101
av_diff_db      = 3.623879e+01
av_diff_mag     = 6.485438e+01
bw_diff         = 7.722065e+04
ugf             = 5.003501e+06
p_ugf           = -8.951775e+01
av_cm_db        = -4.806319e+01
av_cm_mag       = 3.952213e-03
bw_cm           = 7.723316e+04
=== AC Results ===
Av Diff: 36.2388 dB, BW: 77220.6 Hz
UGF: 5.0035E+06 Hz
GBW: 5.0081E+06 Hz
PM: 90.4823 deg
Av CM: -48.0632 dB, BW: 77233.2 Hz
CMRR: 84.302 dB
=== Layout Metrics ===
Total Active MOSFET Area: 2.286E-11 m^2
binary raw file "5t_ota_tb_ol.raw"
ngspice 1 ->

```

Figure 9: Ngspice AC after increasing width

2.3 Common-mode small-signal AC

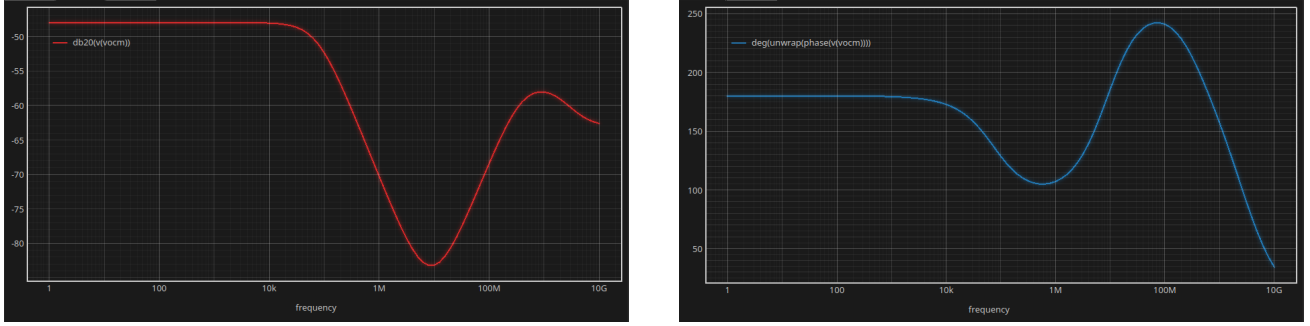


Figure 10: $|A_{v,cm}|$ vs frequency (dB) (left) and $\angle A_{v,cm}$ vs frequency (right)

CM gain of a 5T OTA (matched devices) is approximately:

$$A_{v,cm} \approx -\frac{1}{2 g_{m3} r_{o5}}$$

Using $g_{m3} = 100.96 \mu A/V$ and $r_{o5} = 1/g_{ds5} = 1/1.0389 \mu A/V = 962.6 k\Omega$:

$$A_{v,cm} = -\frac{1}{2(100.96\mu)(962.6k)} = -\frac{1}{194.4} = -5.14 \times 10^{-3} = -45.8 \text{ dB}$$

Table 3: Common-mode AC – hand calc vs sim

Quantity	Hand calc	Simulation
$A_{v,cm}$	-45.8 dB	-47.97 dB
BW_{cm}	76.9 khz	76.1 kHz

2.4 CMRR

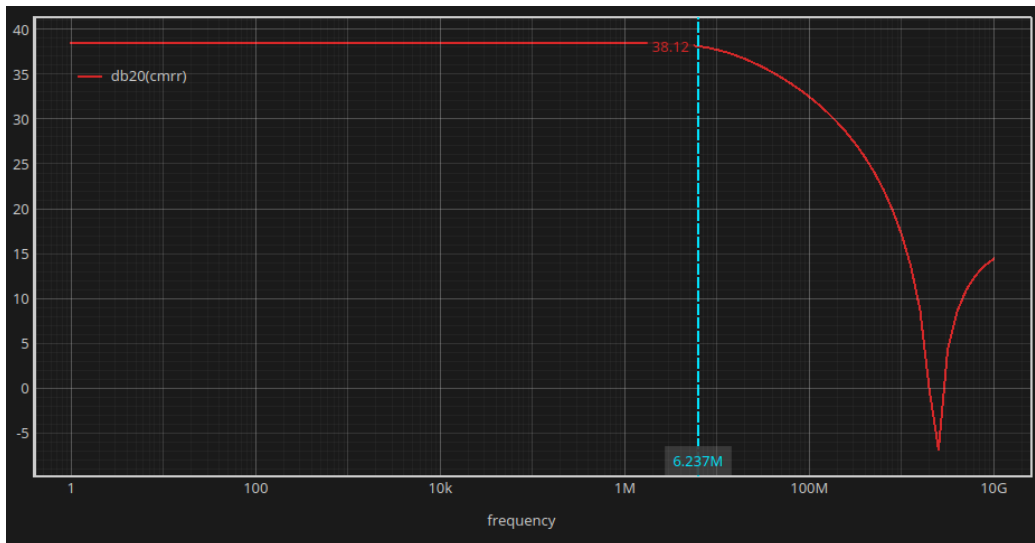


Figure 11: CMRR vs frequency

$$CMRR_{hand} = A_{v,diff}(dB) - A_{v,cm}(dB) = 36.16 - (-45.8) = 81.9 \text{ dB}$$

Simulation gives $CMRR = A_{v,diff_db} - A_{v,cm_db} = 36.13 - (-47.97) = 84.1 \text{ dB}$.

Table 4: CMRR – hand calc vs sim

Quantity	Hand calc	Simulation
$CMRR$	82 dB	84 dB

2.5 Avcm vs VICM (CMIR sweep)

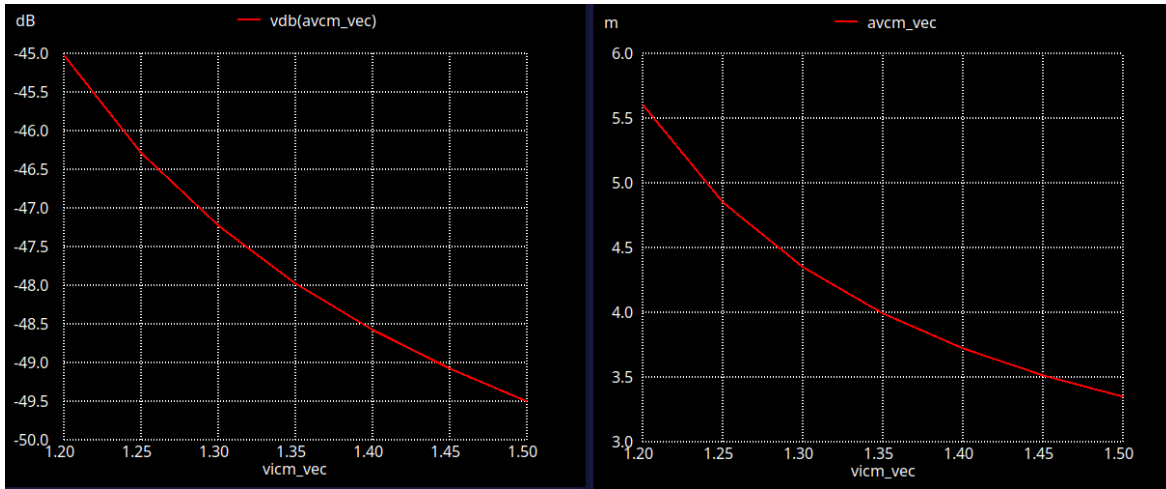


Figure 12: $A_{v,cm}$ (dB, left) and (linear, right) vs VICM, swept across the CMIR

Q: Does Avcm increase or decrease vs VICM? Why?

A: It decreases As VICM goes up, V_{DS} of the tail transistor M5 increases (it goes deeper into saturation), which increases r_{o5} . Since $A_{v,cm} \propto 1/r_{o5}$, a bigger r_{o5} means smaller CM gain, improving CMRR.

2.6 Differential large-signal sweep

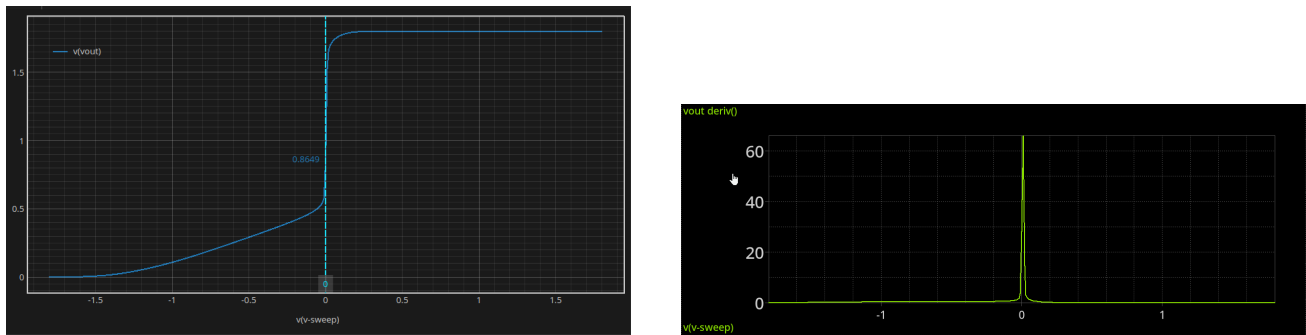


Figure 13: V_{OUT} vs V_{ID} (left) and dV_{OUT}/dV_{ID} vs V_{ID} (right)

Q: What is Vout at VID=0? Why? How does the derivative peak compare to Avd?

A: $V_{OUT}(V_{ID} = 0) = 0.8649$ V as the open-loop DC bias point (0.8653 V).

The derivative dV_{OUT}/dV_{ID} peaks at about 65 V/V at $V_{ID} = 0$, which is the gain $A_{v,diff} = 64.3$ V/V, the derivative of the large-signal DC transfer curve at the origin is the small-signal gain.

2.7 CM large-signal sweep (GBW vs VICM $\rightarrow$ actual CMIR)

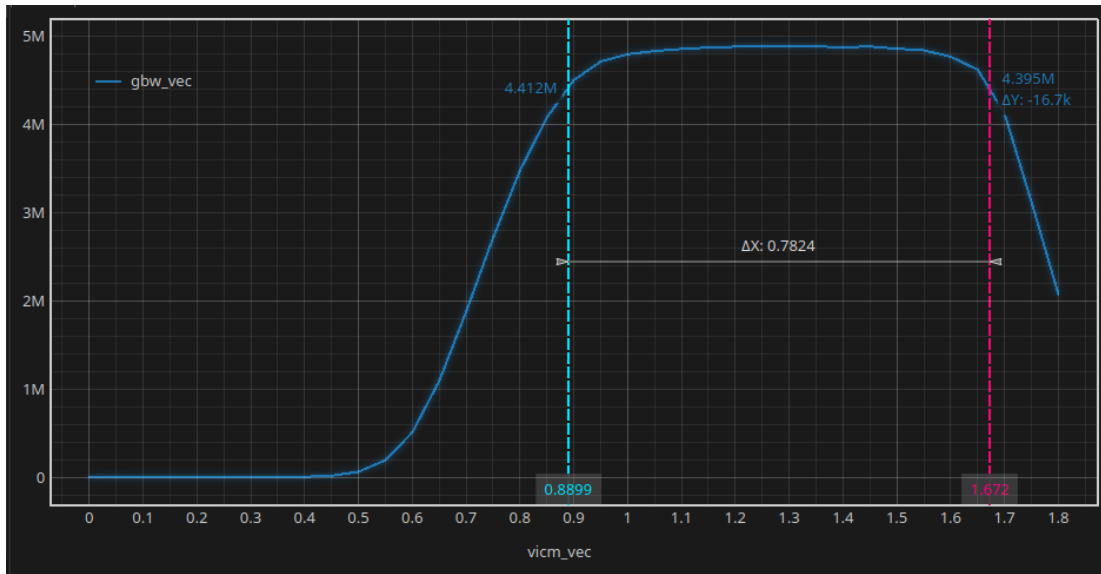


Figure 14: GBW vs VICM

The 90% threshold (≈ 4.41 MHz) is crossed at VICM=0.89 V on the low side and VICM=1.6 V on the high side, giving a simulated CMIR of about [0.89 V, 1.6 V].

3 Part 3: Closed-Loop OTA Simulation (Unity-Gain Buffer)

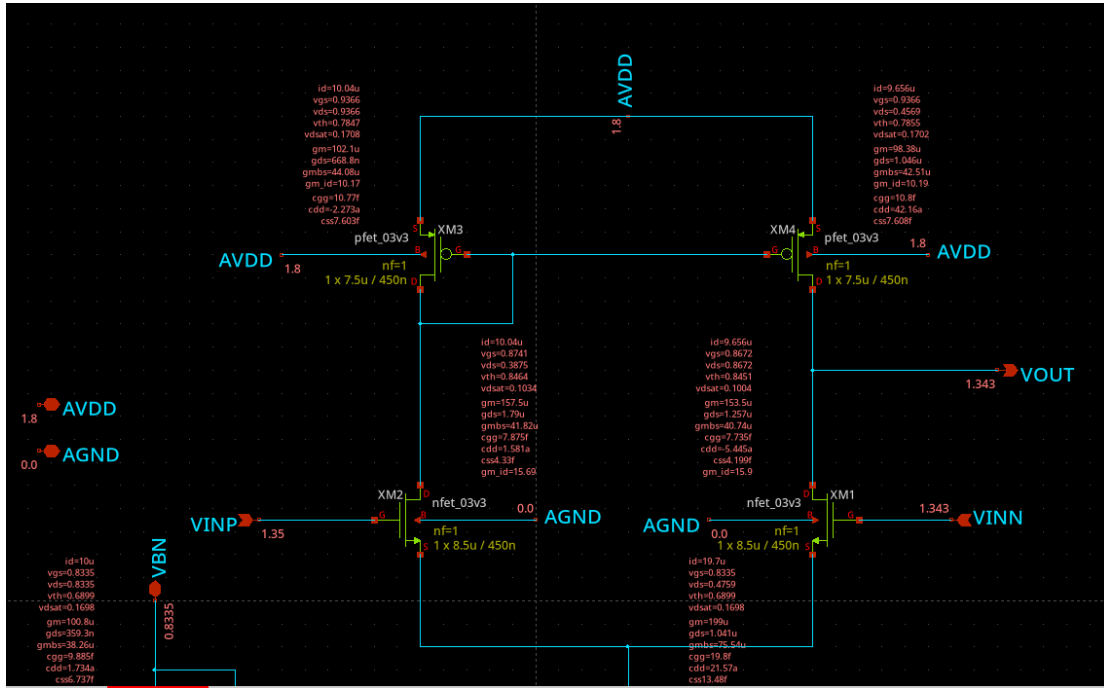


Figure 15: Closed-loop DC OP annotated

Q: Is the current (and gm) in the input pair exactly equal ? Why?

A: No, $I_D(M2) = 10.04 \mu A$ while $I_D(M1) = 9.656 \mu A$ – a difference of about $\Delta I_D \approx 0.38 \mu A$ ($\sim 3.9\%$ mismatch).

in the unity-gain buffer, V_{OUT} follows V_{INP} ($=V_{ICM} = 1.35V$) but only through a *finite*-gain loop, so it can't track perfectly – V_{OUT} settles at 1.343 V, not 1.350 V. That small difference becomes a differential input $V_{id} = V_{INP} - V_{INN} = 1.35 - 1.343 \approx 7 \text{ mV}$. This V_{id} makes the differential pair unbalanced like a mismatch.

3.1 Loop gain (STB analysis)

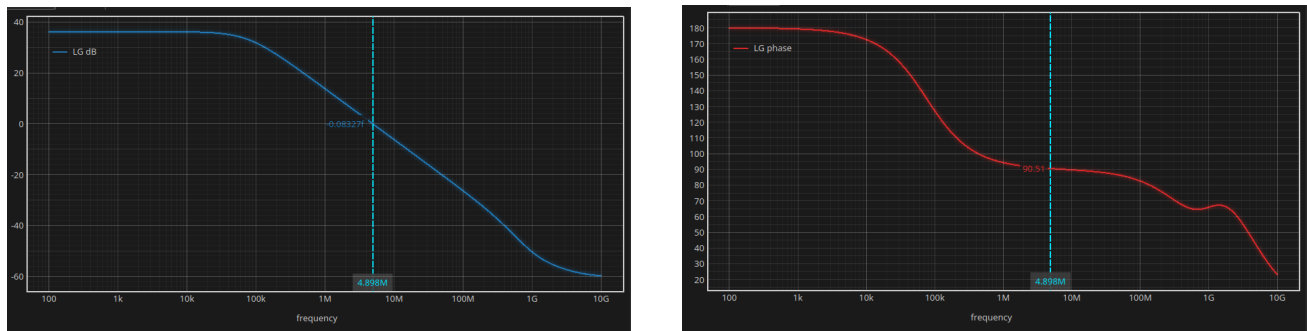


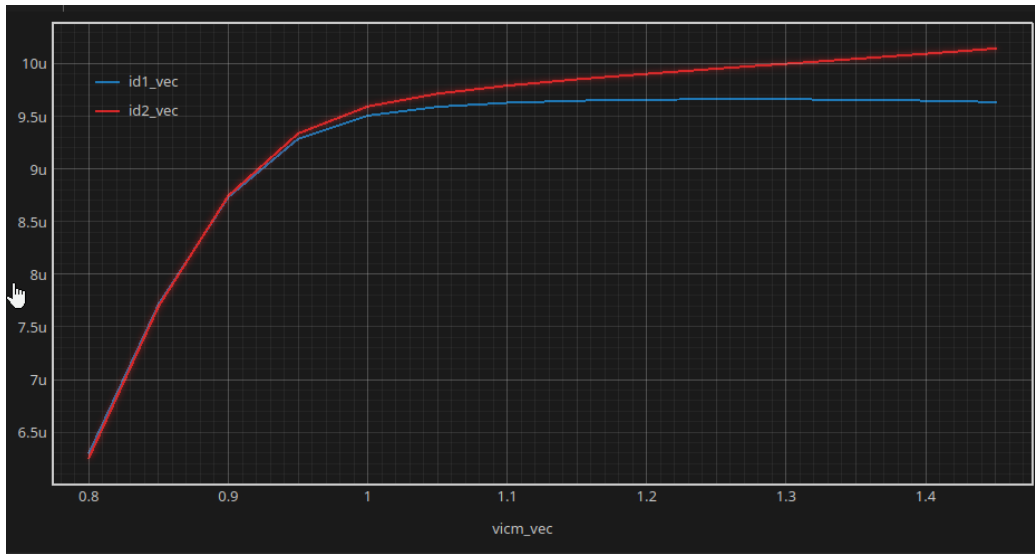
Figure 16: Loop gain magnitude (dB) (left) and Loop gain phase (right)

Table 5: Loop gain vs open-loop diff results

Quantity	Open-loop (Part 2)	Loop gain (Part 3, STB)	Hand calc
DC gain	36.13 dB	≈ 36 dB	36.16 dB
Crossover freq.	UGF = 4.894 MHz	GX = 4.898 MHz	GBW = 4.94 MHz
Phase margin	90.51°	90.51° (at GX)	$\approx 90^\circ$

this is a unity-feedback buffer, so the feedback factor $\beta = 1$, and loop gain $LG = \beta \cdot A_{v,diff} = A_{v,diff}$.

4 Part 4: Effect of Mismatch on CMRR

Figure 17: I_{D1}, I_{D2} vs VICM

For VICM below about 1.0 V, I_{D1} and I_{D2} are rising equally. Above ~ 1.0 V they start to split I_{D2} (red) keeps rising while I_{D1} (blue) is decreasing a little because the addition of them is const. this is because there is a mismatch in the input due to the feedback.

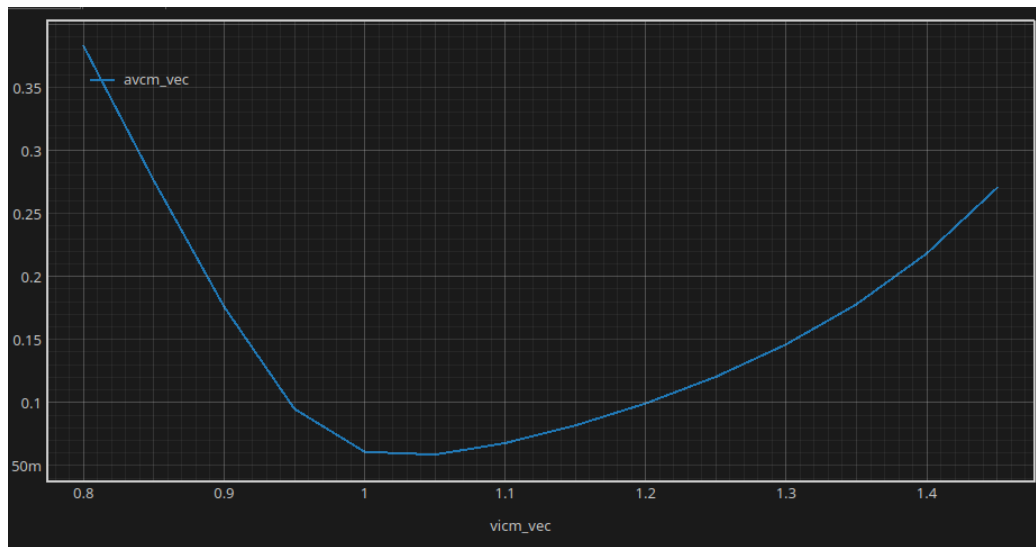


Figure 18: CM gain vs VIN

when V_{Icm} is low the tail mosfet is not in sat so R_{SS} is very small so the A_{Vcm} is large as V_{Icm} increases R_{SS} increases till The common-mode gain will reach its minimum best performance exactly at the point where V_{IN} equals the voltage at the mirror node, because the input pair is perfectly balanced. also due to the feedback there is a systematic mismatch so the current differs.